

Erdős Problem #1033

EP-1033

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $h(n)$ be such that every graph on n vertices with $> n^2/4$ many edges contains a triangle whose vertices have degrees summing to at least $h(n)$. Estimate $h(n)$. In particular, is it true that $h(n) \geq (2(\sqrt{3} - 1) - o(1))n$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>